INCH-POUND
MIL-PRF-5/1J
25 November 2022
SUPERSEDING
MIL-C-5/1H
21 June 2011

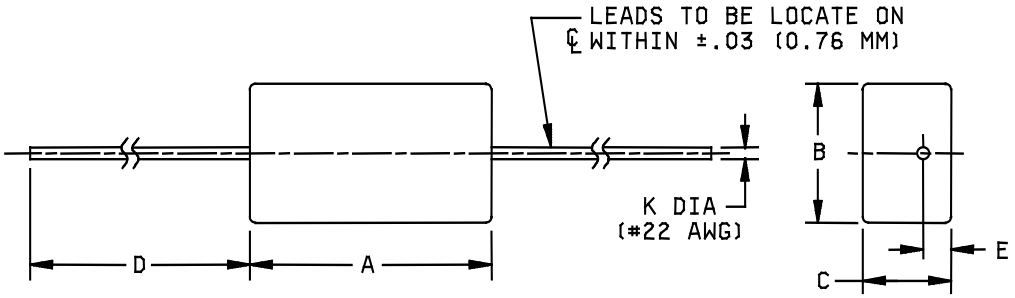
PERFORMANCE SPECIFICATION SHEET

CAPACITOR, FIXED, MICA DIELECTRIC, STYLE CM15

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-5.

INACTIVE FOR NEW DESIGN
after 31 March 1999



DIMENSIONS						
DIM	Inches			mm		
	MIN	NOM	MAX	MIN	NOM	MAX
A	.48	.52	.55	12.19	13.21	13.97
B	.27	.30	.31	6.86	7.62	7.87
C	.14	.19	.22	3.56	4.83	5.59
D	1.12	---	---	28.45	---	---
E	.06	---	---	1.52	---	---
K	.023	.025	.027	.58	.64	.69

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in millimeters and are given for general information only.
3. Approximate weight .05 ounce.
4. Unless otherwise specified, tolerance is ± .02.

FIGURE 1. Style CM15 capacitors.

AMSC N/A
FSC 5910

DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

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REQUIREMENTS:

Dimensions: See figure 1.

Capacitance value: See table I.

Capacitance tolerance: See table I.

Symbol	Capacitance tolerance
D	± 0.5 pF
F	± 1 percent
G	± 2 percent
J	± 5 percent

DC rated voltage: See table I.

Operating temperature range: See table I.

Symbol	Temperature range
N	-55°C to +85°C
O	-55°C to +125°C
P	-55°C to +150°C

Vibration grade: 3 (10 to 2,000 Hz).

Characteristic: See table I.

Symbol	Temperature coefficient P/M/°C	Capacitance drift
C	-200 to + 200	± (0.5 percent +0.1 PF)
E	-20 to + 100	± (0.1 percent +0.1 PF)
F	0 to + 70	± (0.05 percent +0.1 PF)

Immersion: As specified in MIL-PRF-5 except, capacitance change shall not exceed ± 2 percent of nominal value or 1 pf, whichever is greater.

Moisture resistance: As specified in MIL-PRF-5 except, capacitance change shall not exceed ± 2 percent of nominal value or 1 pf, whichever is greater.

Qualification: Not required.

Inspection of product for delivery: Inspection of product for delivery shall consist of group A and group B inspections as specified in MIL-PRF-5.

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TABLE I. Style CM15 capacitors.

Part or identifying number (PIN) 1/	Capacitance (pf)	DC working voltage (volts)	Operating temperature range	Characteristics available for capacitance tolerances			
				± .5 pf (D)	± 1% (F)	± 2% (G)	± 5% (J)
CM15CD010D-3	1	500	N,O,P	C			
CM15CD020D-3	2	500	N,O,P	C			
CM15CD030D-3	3	500	N,O,P	C			
CM15CD050D-3	5	500	N,O,P	C			
CM15CD100J-3	10	500	N,O,P				C
CM15CD110J-3	11	500	N,O,P				C
CM15CD120J-3	12	500	N,O,P				C
CM15CD130J-3	13	500	N,O,P				C
CM15CD150J-3	15	500	N,O,P				C
CM15CD160J-3	16	500	N,O,P				C
CM15CD180J-3	18	500	N,O,P				C
CM15CD200J-3	20	500	N,O,P				C
CM15CD220J-3	22	500	N,O,P				C
CM15CD240J-3	24	500	N,O,P				C
CM15-D270--3	27	500	N,O,P			C, E	C, E
CM15-D300--3	30	500	N,O,P			C, E	C, E
CM15-D330--3	33	500	N,O,P			C, E	C, E
CM15-D360--3	36	500	N,O,P			C, E	C, E
CM15-D390--3	39	500	N,O,P			C, E	C, E
CM15-D430--3	43	500	N,O,P			C, E	C, E
CM15-D470--3	47	500	N,O,P			C, E	C, E
CM15-D510--3	51	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D560--3	56	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D620--3	62	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D680--3	68	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D750--3	75	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D820--3	82	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D910--3	91	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D101--3	100	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D111--3	110	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D121--3	120	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D131--3	130	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D151--3	150	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D161--3	160	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D181--3	180	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D201--3	200	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D221--3	220	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D241--3	240	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D271--3	270	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D301--3	300	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D331--3	330	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D361--3	360	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D391--3	390	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D431--3	430	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D471--3	470	500	N,O,P		C, E, F	C, E, F	C, E, F
CM15-D511--3	510	500	N,O,P		C, E, F	C, E, F	C, E, F

1/ Complete PIN, where applicable, will include additional symbols to indicate characteristic (C, E, or F), capacitance tolerance (D, F, G, or J), and temperature range (N, O, or P).

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Supersession data: Capacitors as described herein with capacitance values ranging from 51 through 510 pf, may be considered as suitable replacements for comparable items formerly covered by MIL-C-5/3C. Recommended replacements are as follows:

MIL-C-5/3C items	Recommended replacement items
CM25E510GN3 thru CM25E511GN3	CM15ED510GN3 thru CM15ED511GN3
CM25E510GO3 thru CM25E511GO3	CM15ED510GO3 thru CM15ED511GO3
CM25E510GP3 thru CM25E511GP3	CM15ED510GP3 thru CM15ED511GP3
CM25E510JN3 thru CM25E511JN3	CM15ED510JN3 thru CM15ED511JN3
CM25E510JO3 thru CM25E511JO3	CM15ED510JO3 thru CM15ED511JO3
CM25E510JP3 thru CM25E511JP3	CM15ED510JP3 thru CM15ED511JP3
CM25F910GN3 thru CM25F511GN3	CM15FD910GN3 thru CM15FD511GN3
CM25F910GO3 thru CM25F511GO3	CM15FD910GO3 thru CM15FD511GO3
CM25F910GP3 thru CM25F511GP3	CM15FD910GP3 thru CM15FD511GP3
CM25F910JN3 thru CM25F511JN3	CM15FD910JN3 thru CM15FD511JN3
CM25F910JO3 thru CM25F511JO3	CM15FD910JO3 thru CM15FD511JO3
CM25F910JP3 thru CM25F511JP3	CM15FD910JP3 thru CM15FD511JP3

Changes from previous issue.. The margins of this specification are marked with vertical lines to indicate where changes from previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army – CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5910-2023-002)

Review activities:
Army - MI

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